agriculture

MDPI

Article

A Study on the Impact of Social Trust on Farmers' Agricultural Waste Resource Utilization Behavior in the Context of Environmental Regulation Policy

Jianhua Wang * and Ling Ma

School of Business, Jiangnan University, Wuxi 214122, China; 6220909008@stu.jiangnan.edu.cn
* Correspondence: wiwth_99@163.com

Abstract: The resource utilization of agricultural waste is an important way to achieve the green transformation of agriculture. Based on micro-survey data from 701 farmers in Jiangsu Province, this paper constructs a binary logistic model to explore how farmers' social trust affects their decisions regarding agricultural waste resource utilization. It also examines the moderating effect of environmental regulation policies between the two. The results show that social trust has a significant positive impact on farmers' agricultural waste resource utilization behavior. Secondly, guiding environmental regulation policies positively moderate the effect of interpersonal trust on farmers' agricultural waste resource utilization behavior, while both guiding and restrictive environmental regulation policies serve as substitute moderators in the relationship between institutional trust and farmers' participation behavior. Finally, heterogeneity analysis shows that, due to differences in household endowments, the impact of interpersonal trust and institutional trust on agricultural waste resource utilization behavior differs across different farmer groups. Therefore, this paper suggests cultivating and enhancing the level of social trust among farmers, thereby promoting their participation in waste resource utilization while strengthening the regulatory role of environmental regulation policies in shaping farmers' behavior.

Keywords: interpersonal trust; institutional trust; environmental regulation policy; agricultural waste resource utilization

check for updates

Academic Editor: Bentivoglio Deborah

Received: 27 February 2025
Revised: 25 March 2025
Accepted: 28 March 2025
Published: 1 April 2025

Citation: Wang, J.; Ma, L. A Study on the Impact of Social Trust on Farmers' Agricultural Waste Resource Utilization Behavior in the Context of Environmental Regulation Policy. *Agriculture* **2025**, *15*, 759. https://doi.org/10.3390/agriculture15070759

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

In the development of China's agricultural modernization and transformation, it is faced with the reality of the difficult problem of high agricultural waste emissions and low levels of resource utilization. According to statistics, China produces nearly 4 billion tons of agricultural waste annually, yet the resource utilization rate of agricultural waste is less than 30% [1]. A large amount of agricultural waste is indiscriminately discharged or treated, causing significant negative impacts on the ecological environment and public health [2]. As an important way to realize the sustainable development of China's agriculture, the development of a circular economy can promote the transformation of agriculture from traditional rough growth to resource-saving and environmentally friendly and realize agricultural production with both economic and ecological benefits [3,4]. The resource utilization of agricultural waste based on the concept of a circular economy has also become a topic through which to solve the problem of agro-ecological environmental governance. Farmers are the main actors of agricultural production and management, and their agricultural waste resource utilization behavior determines the effectiveness of the implementation of waste resource utilization, but also in achieving the green development

of agriculture, the construction of the agricultural power of inherent requirements, and inevitable choices.

The influential factors pertaining to farmers' participation in the resource utilization of agricultural waste have always been of great concern to scholars, and relevant research has focused on the following two aspects. First, from the perspective of the farmer as the main actor, existing studies have analyzed individual and household characteristics, agricultural production and operation characteristics, and the psychological and cognitive traits of farmers. These characteristics include individual and household traits, such as gender, age, education level, and household income [5,6]; agricultural production and operation traits, such as the scale of farming (or breeding), the farmer's part-time work situation, and the proportion of the family labor force [7,8]; and psychological and cognitive traits, which include environmental cognition and behavioral attitudes. The environmental cognition and behavioral attitudes of farmers not only have a direct, positive impact on their agricultural waste resource utilization behavior [9], but also help transform farmers' willingness to resourcefully utilize waste into actual actions [10]. Second, from the perspective of agricultural support, research has found that providing farmers with waste-recycling technologies, services, or an improved infrastructure can change the expected cost-benefit ratio of agricultural waste recycling, which helps to promote farmers' agricultural waste resource utilization [11,12].

The above studies have explored the influential factors pertaining to farmers' waste resource utilization behavior from various perspectives, providing useful references for this paper. Rural ecological environment governance has the key attribute of "collective action", which requires the collective participation of farm households [13]. Social trust, as a core element of social capital, has practical significance in promoting collective action when addressing rural public issues. A good level of social trust plays an important role in promoting individual cooperation, information acquisition, assistance, and technical exchange among farmers and can effectively influence farmers' production behavior [14,15], thereby improving policy implementation efficiency. At the same time, social trust can form an internal soft constraint mechanism, strengthen the behavioral and ethical norms of individual farmers, inhibit the phenomenon of "free-riding" by farmers in ecological environment management, and effectively reduce the implementation cost of environmental regulatory policies. However, current research related to social trust mainly focuses on areas such as farmers' adoption of green production technologies [16], farmland transfer contract choices [17], and farmland leasing [18]. The impact of social trust on farmers' agricultural waste resource utilization behavior remains to be further explored.

Since agricultural waste resource utilization has externality attributes, farmers' participation in agricultural waste resource utilization cannot be separated from the context of environmental regulation policies [19]. Zhang and Fu pointed out that agricultural policies are direct influencing factors for farmers to carry out productive waste resource utilization [20]. Both incentive-based policy support and regulation-based policy constraints can effectively promote farmers' agricultural waste resource utilization [21,22]. Zhao and Zhou found that both the incentive and supervisory roles of reward and punishment policies are effective in mitigating the bias between farmers' recycling willingness and behavior [23]. In addition, some scholars have focused on the interaction effects of environmental regulation policies. For example, Li et al. found that environmental regulation policies and informal institutions have both substitute and complementary effects in promoting farmers' green production behavior [24]. Due to the complexity and heterogeneity of rural public issues, there are differences in the form and extent of environmental regulation policy implementation in rural areas, which leads to differences in farmers' agricultural waste resource

utilization behaviors. Therefore, exploring the differences in farmers' waste resource utilization behavior under different environmental regulation policy contexts has important practical significance.

Based on this, this paper uses research data from 701 agricultural farmers in Jiangsu Province and applies a binary logistic regression model to explore the influence of social trust on farmers' agricultural waste resource utilization behavior and its mechanism of action in the context of environmental regulation policy. Therefore, the possible marginal contributions of this paper are: first, this paper focuses on the micro level to explore the important role of social trust in the decision-making of agricultural waste resource utilization behavior of farm households, and enriches the research on the influencing factors of farm households' participation in agricultural waste resource utilization; second, taking into account the heterogeneous characteristics of farm households in terms of household endowments, the differential impact of social trust across different groups of farm households is further explored to provide empirical evidence for differential policy-making; third, this paper analyzes the moderating effects of different types of environmental regulatory policies in social trust farmer behavior, enriching the research on the mechanisms influencing the behavior of agricultural waste resource utilization by farmers.

2. Theoretical Framework and Hypotheses

2.1. The Impact of Social Trust on Farmers' Agricultural Waste Resource Utilization Behavior

As a core element of informal institutions [25], social trust refers to the level of expected trustworthiness that social members within a specific region form toward other individuals through social interactions [26]. Social trust can effectively reduce the information asymmetry faced by farmers when making decisions on the utilization of agricultural waste resources by enhancing the information sharing among farmers and broadening the information access channels, thus reducing the information access cost and cooperative transaction cost of the utilization of agricultural waste resources, establishing a multi-body cooperation mechanism between farmers and farmers, farmers and village organizations, and farmers and government departments, and improving the efficiency of rural ecological environment management, thus realizing the common governance of the rural ecological environment. The efficiency of ecological environment management can be improved, thus realizing the common governance of rural ecological environment. Research indicates that social trust encompasses both interpersonal trust and institutional trust, and helps enhance public willingness to participate in environmental governance and promotes the collaborative participation of multiple stakeholders in environmental governance actions [27]. On one hand, interpersonal trust is based on emotional bonds between individuals, commonly occurring within rural communities, and mainly manifests as trust toward friends, family, and neighboring farmers, with varying levels of trust. Farmers as "social people", through stable social interaction to promote the formation of interpersonal trust, coupled with the emotional identity formed by kinship, geographic location, for the establishment of a good basis for cooperation between farmers. On the other hand, institutional trust is rooted in formal policies, regulations, and other institutional environments, which can strengthen the communication between the government and farmers, and form a soft constraint on farmers, which can help to drive farmers to maintain agro-ecological environments and shift to the utilization of waste resources. Existing research indicates that both interpersonal trust and institutional trust significantly promote farmers' behavioral decisions regarding agricultural waste resource utilization [28]. Based on this, the following hypotheses are proposed:

Hypothesis 1 (H1). *Interpersonal trust has a positive impact on farmers' agricultural waste resource utilization behavior.*

Hypothesis 2 (H2). *Institutional trust has a positive impact on farmers' agricultural waste resource utilization behavior.*

2.2. The Moderating Role of Environmental Regulation Policies in the Impact of Social Trust on Farmers' Agricultural Waste Recycling Behavior

Agricultural waste resource utilization exhibits typical environmental externalities. The direct emission of agricultural waste results in negative environmental externalities, whereas its resource utilization generates positive environmental externalities. When farmers make decisions regarding agricultural waste resource utilization, there is a mismatch between individual marginal benefits and social marginal benefits [29]. For the purpose of maximizing economic benefits, farmers are unlikely to voluntarily choose agricultural waste resource utilization. According to externality theory, to better address the negative environmental externalities in the agricultural waste disposal process, the government must intervene in farmers' agricultural waste resource utilization behaviors through policy measures. This intervention would help balance individual marginal benefits with social marginal benefits [30]. Therefore, externality theory serves as a crucial theoretical foundation for explaining how environmental regulation policies guide or constrain farmers' production behaviors. Clarifying how various types of environmental regulation policies promote farmers' behavior within the context of environmental governance policies forms the research motivation of this paper.

Existing research indicates that the current environmental regulation policies promoting agricultural waste resource utilization can be classified into three types: guiding, constraining, and incentivizing regulations [8]. These policies intervene in farmers' production behaviors to achieve the goals of rural ecological environment governance. Specifically, the government conducts publicity and training on the utilization of agricultural waste resources for farmers through guided environmental regulation policies [31], increases communication between farmers and peers and village cadres, and improves the level of social trust of farmers, so as to guide farmers to devote themselves to the production practice of agricultural waste resource utilization. Zhu et al. pointed out that there are similarities between constrained environmental regulation and incentive-based environmental regulation in the mechanism of action [32], both of which affect the level of social trust of farmers through the construction of reward and punishment mechanisms. In the context of the constrained environmental regulation policy, the farmer group may be subject to more external norms. In order to avoid being punished, the farmer's behavioral decision-making on waste resourcing will seek more cooperation, which results in a higher level of trust in the relevant actors. In the case of incentive-based environmental regulation policy, when the government provides economic incentives for farmers' waste resource treatment behavior, farmers' social trust is transformed into the expectation of policy stability, which enhances the positive influence of social trust on agricultural waste resource treatment behavior. Based on this, the following hypothesis is proposed:

Hypothesis 3 (H3). *Environmental regulation policies have a positive moderating effect on the relationship between social trust and farmers' agricultural waste resource utilization behavior.*

Based on the relevant research findings both domestically and internationally, as well as the above hypotheses, this paper constructs the theoretical model (see Figure 1).

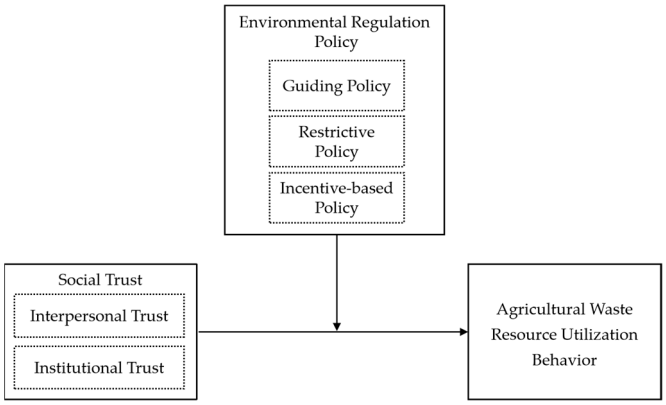

Figure 1. Theoretical model of social trust's influence on farmers' agricultural waste recycling behavior.

3. Model, Data and Variables

3.1. Research Methodology

This paper primarily investigates the impact of social trust on farmers' agricultural waste resource utilization behavior. Therefore, the explanatory variables in this paper are agricultural waste resource utilization behavior, including participation and non-participation, which are dichotomous variables and are empirically analyzed by using a binary logistic regression model. The model transforms the linear prediction value into probability value through logit transformation, which ensures that the prediction result is strictly limited to the interval of [0, 1], so as to clarify the tendency of production decision-making behavior of farmers; at the same time, the logistic regression model has the interpretability of economics and ecology in explaining the coefficients, which is in line with the analytical needs of this paper. Based on existing research, an influencing factors model for farmers' agricultural waste resource utilization behavior has been constructed. The specific form of the model is presented in Equation (1):

$$y_i = \beta_0 + \beta_1 Trust_1 + \beta_2 Trust_2 + \beta_3 Control + \varepsilon \tag{1}$$

In Equation (1), when farmers engage in agricultural waste resource utilization behavior, $y_i = 1$; otherwise $y_i = 0$. Here, $Trust_1$ represents interpersonal trust, $Trust_2$ represents institutional trust, and $Control$ represents control variables, including individual characteristics, family management characteristics, external characteristics, environmental perception characteristics, and regional dummy variables. β_0 is the constant term, β_1, β_2, and β_3 are the regression coefficients for the above variables, and ε is the random disturbance term. Based on this, the functional form is presented in Equation (2):

$$\ln\left(\frac{p_i}{1-p_i}\right) = \partial + \sum_{j=1}^{n} \beta_j x_j + \varepsilon \tag{2}$$

In Equation (2), x_j represents the j factor influencing farmers' implementation of agricultural waste resource utilization behavior, ∂ is the constant term, β_j is the regression coefficient, and ε is the error term. $\frac{p_i}{1-p_i}$ represents the odds ratio of a specific farmer engaging in agricultural waste resource utilization compared to not engaging in such utilization.

3.2. Data Sources and Sample Description

The data used in this paper come from the field research conducted by the research group from October 2022 to January 2023 in the agricultural planting areas of Jiangsu Province, and in July 2023, supplementary research was conducted. The main reasons for choosing Jiangsu Province as the sample province are as follows: first, as one of the

13 main grain-producing areas in China, Jiangsu Province faces the reality of the significant increase in agricultural waste and the management of agricultural surface pollution in the process of increasing grain production; second, Jiangsu Province pays great attention to the problem of agricultural waste treatment and actively responds to the national call for the resource utilization of agricultural waste. Based on the combination of stratified sampling and random sampling, one representative city was selected as the sample city in North Jiangsu, Central Jiangsu, and South Jiangsu, namely Suqian City, Taizhou City, and Wuxi City. Firstly, the regional agricultural output value in the three sample cities is ranked, and one county (city and district) is selected, respectively; secondly, the number of agricultural growers is used as the basis of classification, and two sample townships (towns) are randomly selected among the top 50% of the townships (towns); again, the number of agricultural growers is used as the basis of classification, and two of townships (towns) are randomly selected as the sample townships (towns); lastly, the number of agricultural growers is randomly 20–25 farming households were selected to conduct one-on-one interviews with their family agricultural production and management decision makers. According to the theme and purpose of the study, after eliminating invalid questionnaires, 701 valid questionnaires were finally used for the study.

The survey results show that the majority of respondents are male, comprising 60.90% of the sample. Additionally, 73.5% of respondents are over the age of 50, which aligns with the current demographic characteristics of rural Jiangsu, suggesting that the sample is representative. Furthermore, 69.0% of respondents have an education level no higher than middle school, which may limit their awareness and understanding of agricultural waste resource utilization. In terms of family and agricultural production characteristics, the majority of respondents have 2 to 4 household members engaged in labor, accounting for 65.8%. 68.6% of respondents reported that agricultural income constitutes less than 40% of their total household income. 88.9% of respondents are ordinary farmers, while only 8.2% are professional large-scale farmers or family farm owners. Regarding environmental awareness, 38.7% of respondents agree that agricultural waste resource utilization benefits ecological environmental protection. Similarly, less than 40% of respondents are optimistic about the future prospects of agricultural waste resource utilization.

3.3. Variable Selection

3.3.1. Dependent Variable

The dependent variable in this study is agricultural waste resource utilization behavior. Agricultural waste resource utilization refers to the direct or regenerated use of agricultural waste such as livestock manure, crop straw, and pesticide packaging materials through processes such as feed processing, fertilization, and energy generation [33]. Therefore, in the actual agricultural production process, if farmers recycle pesticide packaging (e.g., pesticide bottles, fertilizer bags, etc.), agricultural film (plastic film used to cover agricultural fields, greenhouses, greenhouses, etc.), etc., or reuse crop stalks (e.g., wheat stalks, rice stalks, maize stalks, etc.), this indicates that farmers have participated in the resource utilization of agricultural waste, and the explanatory variable is assigned a value of 1, while the opposite value assigned is 0.

3.3.2. Explanatory Variables

Drawing on the work of He et al., this study divides social trust into interpersonal trust and institutional trust [28]. Specifically, the scale of interpersonal trust includes 2 items: "I trust my relatives very much, and they influence my agricultural waste resource utilization behavior" and "I trust the surrounding farmers very much, and they influence my agricultural waste resource utilization behavior". The scale of institutional trust includes 2 items:

"I trust the village officials very much, and they influence my agricultural waste resource utilization behavior" and "I have great trust in policies, and I believe the government can promote the regulation of agricultural waste pollution". The scale was measured with a 5-point Likert scale ranging from "strongly disagree" to "strongly agree". The reliability value was determined as 0.895 and 0.869. The above four questions are all rated using a 5-point Likert scale. Interpersonal trust and institutional trust are measured by calculating the average value of the two respective indicators [17].

3.3.3. Moderating Variables

The scale of environmental regulation policy is adapted from Wang et al. and includes 3 items with 5-point scales [8]: "Government promotion and training on agricultural waste resource utilization (from never to always)", "Strong regulatory pressure from environmental protection agencies during agricultural waste resource utilization (from strongly disagree to strongly agree)" and "Government economic incentives for agricultural waste resource utilization (from never to always)". The scale was measured with a 5-point Likert scale. The Cronbach alpha coefficient value was found to be 0.682, which is acceptable.

This paper used AMOS 26 to test the validity, and the results are shown in Table 1. The AVE values are both greater than 0.5, and the CRs are greater than 0.8, indicating that the convergent validity is ideal. In addition, the square root of AVE on the diagonal was greater than the correlation coefficients between the variables, which suggests better discriminant validity of the scale.

Table 1. Results of validity.

	AVE	CR	Interpersonal Trust	Institutional Trust	Environmental Regulation Policy
interpersonal trust	0.810	0.895	0.900		
institutional trust	0.847	0.917	−0.055	0.920	
environmental regulation policy	0.550	0.826	0.169 ***	0.239 ***	0.742

Note: *** represents significance level 1%. Diagonal values show $\sqrt{AVE}$ for each variable.

3.3.4. Control Variables

Drawing on existing studies [34], this paper selects control variables from two dimensions: intrinsic endowment and external environment. Specifically, the control variables are as follows: (1) individual characteristics, including the respondent's gender, age, and education level; (2) family business characteristics, including the number of family laborers, the proportion of agricultural income, and the mode of production and operation; (3) external characteristics, including whether there are fixed agricultural waste recycling points nearby and their respective distances; (4) environmental perception characteristics, mainly concerning the perceived value and future prospects of agricultural waste resource utilization; (5) regional dummy variables.

The meanings of variables and their descriptive statistics are presented in Table 2.

Table 2. The meanings of variables and their descriptive statistics.

Variable Type	Variables	Definition	Mean	Std. Dev.
Dependent Variable	agricultural waste resource utilization behavior	Has the respondent participated in the resource utilization of agricultural waste: Yes = 1; no = 0	0.725	0.447

Table 2. *Cont.*

Variable Type	Variables	Definition	Mean	Std. Dev.
Explanatory Variables	interpersonal trust	I trust my relatives and the surrounding farmers very much, and they influence my agricultural waste resource utilization behavior: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.606	0.923
	institutional trust	"I trust the village officials very much, and they influence my agricultural waste resource utilization behavior" and "I have great trust in policies, and I believe the government can promote the regulation of agricultural waste pollution": strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.526	0.693
Moderating Variables	guiding policy	Government promotion and technical training on agricultural waste resource utilization: never = 1; occasionally = 2; generally = 3; frequently = 4; always = 5	2.769	1.065
	restrictive policy	Strong regulatory pressure from environmental protection agencies during agricultural waste resource utilization: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.047	0.746
	incentive-based policy	Government economic incentives for agricultural waste resource utilization: never = 1; occasionally = 2; generally = 3; frequently = 4; always = 5	2.850	1.172
Control Variables	gender	Male = 1; female = 0	0.609	0.488
	age	18–29 = 1; 30–39 = 2; 40–49 = 3; 50–59 = 4; above 60 = 5	4.024	0.964
	education level	Middle school and below = 1; vocational/high school = 2; undergraduate = 3; graduate and above = 4	1.438	0.712
	numbers of family labor	Actual number	3.736	2.023
	agricultural income proportion	0–20% = 1; 21–40% = 2; 41–60% = 3; 61–80% = 4; 81–100% = 5	2.093	0.235
	mode of production	Ordinary farmer = 1; professional household = 2; family farm = 3; farmers' cooperative = 4; agricultural enterprise = 5	1.198	0.632

Table 2. *Cont.*

Variable Type	Variables	Definition	Mean	Std. Dev.
Control Variables	fixed agricultural waste recycling point	There are local fixed agricultural waste recycling points: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.458	0.977
	distance to fixed agricultural waste recycling point	The agricultural waste recycling point is nearby: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.094	1.055
	perceived value of agricultural waste resource utilization	Agricultural waste resource utilization is beneficial for improving the rural ecological environment: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.735	0.808
	expectations for agricultural waste resource utilization	I have an optimistic expectation for the future of agricultural waste resource utilization: strongly disagree = 1; disagree = 2; generally agree = 3; agree = 4; strongly agree = 5	3.462	0.741
	regional dummy variables	Suqian = 1; other = 0 Taizhou = 1; other = 0 Wuxi = 1; other = 0	0.265 0.310 0.425	0.442 0.463 0.495

4. Results

4.1. Correlation Analysis

Table 3 presents the results of the correlation analysis. The correlation coefficients of interpersonal trust, institutional trust, and the utilization of agricultural waste resources are 0.126 and 0.168, respectively, indicating that there is a positive correlation between the independent variables and the dependent variable; the two-sided significance level is at the level of 0.01, which indicates that as the level of trust increases, the level of the utilization of agricultural waste resources also increases.

Table 3. The results of the correlation analysis.

Variables	Agricultural Waste Resource Utilization Behavior	Interpersonal Trust	Institutional Trust	Guiding Policy	Restrictive Policy	Incentive-Based Policy
agricultural waste resource utilization behavior	1					
interpersonal trust	0.126 **	1				
institutional trust	0.168 **	0.042	1			
guiding policy	−0.02	0.143 **	0.324 **	1		
restrictive policy	−0.001	0.186 **	0.325 **	0.491 **	1	
incentive-based policy	−0.054	0.147 **	0.114 **	0.424 **	0.340 **	1

Note: ** represent significance level of 0.01.

4.2. Basic Regression

Table 4 presents the baseline regression results. Model (1) includes only the two variables reflecting social trust, while model (2) through (5) progressively add control variables corresponding to farmers' individual characteristics, family operation characteristics, external characteristics, and environmental perception characteristics. The pseudo R^2 of the regression equation gradually increases from 0.087 to 0.160, indicating that the explanatory power of the model strengthens progressively. Moreover, the marginal effects of each variable are calculated based on the results of model (5) to examine the extent to which each variable affects farmers' agricultural waste resource utilization behavior. Since the pseudo R^2 in model (5) is the highest among the five regression models, indicating the strongest explanatory power, this paper primarily discusses the estimation results of model (5), as presented in Table 2.

Table 4. Regression results of the models of the impact of social trust on farmers' agricultural waste resource utilization behavior.

Variables	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Marginal Effects (Based on Model (5))
interpersonal trust	0.256 *** (0.099)	0.260 ** (0.102)	0.250 ** (0.104)	0.247 ** (0.104)	0.199 * (0.109)	0.032 * (0.018)
institutional trust	0.871 *** (0.147)	0.864 *** (0.150)	0.845 *** (0.152)	0.822 *** (0.167)	0.620 *** (0.173)	0.101 *** (0.027)
gender		0.886 *** (0.189)	0.892 *** (0.193)	0.922 *** (0.195)	0.899 *** (0.198)	0.147 *** (0.031)
age		−0.026 (0.116)	−0.048 (0.118)	−0.062 (0.120)	−0.053 (0.122)	−0.009 (0.020)
education level		0.054 (0.164)	0.016 (0.167)	−0.007 (0.168)	−0.094 (0.174)	−0.015 (0.028)
numbers of family labor			0.115 (0.073)	0.109 (0.073)	0.101 (0.072)	0.016 (0.012)
agricultural income proportion			−0.124 (0.081)	−0.144 * (0.082)	−0.160 * (0.084)	−0.026 * (0.014)
mode of production			0.543 *** (0.203)	0.568 *** (0.205)	0.438 ** (0.203)	0.071 ** (0.031)
fixed agricultural waste recycling point				0.209 (0.129)	0.005 (0.139)	0.001 (0.023)
distance to fixed agricultural waste recycling point				−0.157 (0.117)	−0.216 * (0.123)	−0.035 * (0.020)
perceived value of agricultural waste resource utilization					0.454 *** (0.162)	0.074 *** (0.026)
expectations for agricultural waste resource utilization					0.339 * (0.176)	0.055 * (0.029)

Table 4. *Cont.*

Variables	Model (1)	Model (2)	Model (3)	Model (4)	Model (5)	Marginal Effects (Based on Model (5))
regional dummy variables	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Prob > chi^2	0.000	0.000	0.000	0.000	0.000	
Pseudo R^2	0.087	0.115	0.134	0.138	0.160	
log likelihood	−376.544	−365.240	−357.388	−355.773	−346.346	
numbers of observation	701	701	701	701	701	

Note: *, **, and *** respectively represent significance levels of 10%, 5%, and 1%. Robust standard errors are in parentheses.

4.2.1. Impact of Explanatory Variables

The results of model (5) in Table 4 show that, first, interpersonal trust is positively significant at the 10% level, with a marginal effect of 3.2%, confirming Hypothesis H1. This implies that, under the same conditions, for each level increase in farmers' trust towards family members, relatives, and neighboring farmers, the probability of their participation in agricultural waste resource utilization increases by 3.2 percentage points. This may be because "trust among acquaintances" is a key orientation in traditional Chinese social relationships, where kinship and neighborhood ties are crucial resources for farmers. These relationships facilitate long-term interactions and cooperation among farmers [35], effectively reducing information asymmetry and transaction costs between them. The successful implementation of agricultural waste resource utilization requires substantial investments of time, economic resources, and other costs. However, mutual trust among family members and neighbors not only provides cooperative support but also enables shared risk, thus promoting farmers' involvement in agricultural waste resource utilization.

Second, institutional trust is statistically significant at the 1% level, with a positive coefficient and a marginal effect of 10.1%, confirming Hypothesis H2. This means that, under the same conditions, for each level increase in farmers' trust towards village leaders and the government, the probability of their participation in agricultural waste resource utilization increases by 10.1 percentage points. This suggests that farmers with higher levels of trust in village leaders and the government are more likely to engage in agricultural waste resource utilization. A possible explanation is that village leaders, as endogenous forces in rural social governance, are embedded in rural social networks and have a strong mobilizing ability to encourage farmers' participation in agricultural waste resource utilization [36]. Additionally, government intervention, as an external constraint, plays an important role in promoting rural agricultural development. An increase in farmers' trust towards village leaders or the government to some extent reflects their motivation to comply with institutional constraints [37], thereby facilitating farmers' involvement in agricultural waste resource utilization.

Third, based on the results of the marginal effects, the marginal effect of institutional trust is 6.9 percentage points higher than that of interpersonal trust, indicating that, compared to interpersonal trust, institutional trust has a more significant impact on promoting farmers' agricultural waste resource utilization behavior. The possible reason is that the utilization of agricultural waste resources is a typical environmental positive externality activity, in order to achieve the rural ecological environmental governance, the collective participation of farmers is needed [38]. In order to avoid the phenomenon of "free-riding"

in the participation of farmers in environmental management, it is necessary to regulate the behavior of farmers by external constraints such as village cadres and government.

4.2.2. Impact of Control Variables

Table 4 shows that, in terms of individual characteristics, the estimated coefficient for gender is significantly negative at the 1% level, while age and education level do not have a significant impact on farmers' agricultural waste resource utilization behavior. In terms of household business characteristics, the proportion of agricultural income passed the significance test with a negative regression coefficient. A possible explanation for this result is that a lower proportion of agricultural income indicates a more diversified livelihood structure for agricultural producers, which in turn suggests that farmers may have the economic foundation necessary to facilitate the conversion of agricultural waste into resource utilization. The production and management method variable is significantly positive at the 5% level, indicating that as agricultural production methods become more large-scale and specialized, farmers are more likely to engage in agricultural waste resource utilization behavior. In terms of external characteristics, fixed agricultural waste collection points did not pass the significance test, which is inconsistent with the expected result. A possible explanation for this is that the disposal of agricultural waste is a frequent activity for agricultural producers, which is largely influenced by personal habits and convenience factors [39]. Farmers tend to choose low-cost, non-resource utilization methods for handling agricultural waste. However, the estimated coefficient for the distance to fixed agricultural waste collection points is significantly negative at the 10% level, indicating that the closer the fixed agricultural waste collection point is to the farmer, the more likely the farmer is to engage in waste resource utilization. In terms of environmental perception, farmers' recognition of the value and prospects of agricultural waste resource utilization both significantly positively affect their behavioral decisions. This may be because farmers' recognition of the value and prospects of agricultural waste resource utilization reflects their level of understanding [28]. The more farmers understand the benefits of agricultural waste resource utilization and the potential for future implementation, the more likely they are to make decisions regarding agricultural waste resource utilization.

4.3. Moderating Effect Test

This paper examines the moderating effect of environmental regulation policies as a contextual factor using the following specific method: First, the mean values of the three moderating variables—guiding, constraining, and incentive-based environmental regulation policies—are used as grouping criteria. Farmers are then divided into two groups: below the mean and above the mean [40]. Next, this paper treats farmers' agricultural waste resource utilization behavior as the dependent variable, with interpersonal trust and institutional trust as independent variables. Binary logistic regression is then conducted separately on the two groups. Finally, the method of intergroup coefficient difference test [16] is used to compare the significance level and coefficient size of the two groups under different environmental regulation policy situations, thus analyzing the regulatory role of environmental regulation policy. If the direction of regulation is positive, it is considered that social trust and environmental regulation policy have a complementary relationship, and vice versa [41], i.e., there is an overlap in the guiding or restraining function of social trust and environmental regulation policy in regulating the participation of farmers in the utilization of waste resources and thus showing a substitute relationship. The regression results are shown in Table 5.

Table 5. The results for the moderating effects of environmental regulation policies.

Variables	Guiding Policy		Restrictive Policy		Incentive-Based Policy	
	Below Mean Group	Above Mean Group	Below Mean Group	Above Mean Group	Below Mean Group	Above Mean Group
interpersonal trust	0.070 (0.125)	0.704 ** (0.280)	0.196 (0.124)	0.510 (0.311)	0.353 * (0.210)	0.105 (0.138)
institutional trust	0.728 *** (0.197)	−0.508 (0.601)	0.689 *** (0.195)	1.140 ** (0.554)	0.809 ** (0.361)	0.552 *** (0.212)
control variables	controlled	controlled	controlled	controlled	controlled	controlled
Prob > chi^2	0.000	0.000	0.000	0.000	0.000	0.000
Pseudo R^2	0.180	0.232	0.177	0.303	0.312	0.217
log likelihood	−266.991	−64.606	−271.181	−57.985	−94.065	−213.302
numbers of observation	535	166	560	141	260	441
empirical *p*-value	interpersonal trust 0.070	institutional trust 0.073	interpersonal trust 0.384	institutional trust 0.051	interpersonal trust 0.363	institutional trust 0.547

Note: *, **, and *** respectively represent significance levels of 10%, 5%, and 1%. Robust standard errors are in parentheses. The empirical *p*-value is the result of comparing the significance of coefficient differences between two sample groups using an unrelated models test.

The results show that in the context of guiding environmental regulation policies, the empirical *p*-values for the coefficient differences between the interpersonal trust and institutional trust groups are both significant, indicating that guiding environmental regulation policies have a moderating effect on the influence of social trust on farmers' agricultural waste recycling behaviors. However, the significance level of the coefficient for interpersonal trust is higher in the above-mean group, while the significance level for institutional trust is higher in the below-mean group, indicating that the role of guiding environmental regulation policies differs. There is a positive moderating effect in the relationship between interpersonal trust and farmers' agricultural waste recycling behaviors, but a negative moderating effect in the relationship between institutional trust and these behaviors, indicating a substitute moderating effect. This suggests that, on the one hand, in the context of guided environmental regulation policy with greater strength, it can promote farmers with higher interpersonal trust level to practice agricultural waste resource utilization; on the other hand, guided environmental regulation policy forms endogenous driving force of farmers through education and propaganda, while institutional trust enhances the initiative of farmers to participate in waste resource utilization and thus guides their behavioral decision-making, therefore, the two show an overlap in the guidance function and present an alternative relationship [42].

In the context of constraint-based environmental regulation policies, the empirical *p*-value for the coefficient differences between the interpersonal trust groups is not significant. However, the empirical *p*-value for the coefficient differences between the institutional trust groups is significant, and compared to situations with stronger constraint-based environmental regulation policies, the significance level of institutional trust is higher in the below-mean group, indicating a negative moderating effect of constraint-based environmental regulation policies in the institutional trust—farmer participation behavior relationship, meaning that constraint-based environmental regulation policies have a substitutive moderating effect in this relationship. A possible explanation is that farmers' trust

in village cadres or the government is rooted in daily life or agricultural production, which to some extent can form informal institutional soft constraints within rural communities and restrict irregular behaviors in farmers' production activities through external pressure. In contrast, constraint-based environmental regulation policies impose accountability or punishment on non-compliant farmers, meaning there is an overlap in the constraining functions of farmers' institutional trust and constraint-based environmental regulation policies. As a result, they have a substitutive effect on the farmers' agricultural waste recycling behaviors.

Incentive-based environmental regulation policies do not have a moderating effect on the influence of interpersonal trust or institutional trust on farmers' agricultural waste recycling behaviors, which is similar to the research conclusions of Wang et al. [8]. The above research results indicate that Hypothesis H3 is partially supported (see Figure 2). Possible explanations are that, on the one hand, there is a cognitive bias in farmers' perception of the incentive-based environmental regulation policy, interpreting it as a short-term administrative task rather than a continuous norm, resulting in a limited policy regulation; on the other hand, due to the lack of governmental publicity on the incentive policy for the resource utilization of agricultural wastes, coupled with the fact that the economic incentives provided by the government may not have adequately met the expectations and needs of the farmers, the policy perception of the farmers has not been transformed into policy trust, thus inhibiting the resource utilization behavior of agricultural waste.

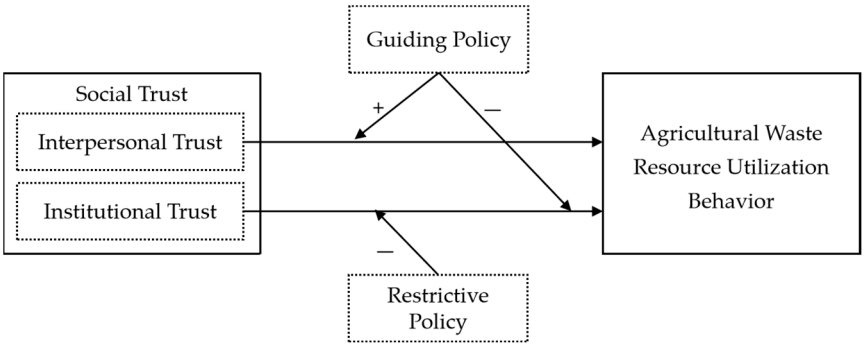

Figure 2. Moderation effect test results.

4.4. Robustness Tests

To test the robustness of the baseline regression results, this study conducted robustness checks using three methods: model substitution, extreme value treatment, and sample restriction. The results are shown in Table 6.

(1) Model substitution. Considering that the dependent variable in this study is a binary variable, the Probit model is used to re-examine the impact of social trust on farmers' agricultural waste recycling behaviors. The results obtained are consistent with the baseline regression. (2) Extreme value treatment. During the micro-survey, farmers might "overestimate" or "underestimate" their true intentions out of politeness or for strategic reasons, leading to extreme values at both ends of the sample [24]. Therefore, this study uses the Winsorization method to smooth extreme values in the top and bottom 5% of the sample data, and then re-conducts binary logistic regression analysis. (3) Sample restriction. Considering that farmers of different ages have differences in learning ability and cognitive levels, and this difference might affect their agricultural waste recycling behaviors, as well as the disadvantages faced by the elderly in agricultural production activities, this study excludes the elderly sample (i.e., respondents aged 60 or above), based on internationally recognized standards for the aging population and previous studies [43]. The regression analysis is then re-conducted, with the results shown in column (3). The

regression estimation results in Table 4 are consistent with those in Table 2, indicating that the empirical analysis results of the baseline regression are robust.

Table 6. The results of the robustness tests.

Variables	Model Substitution (1)	Extreme Value Treatment (2)	Sample Restriction (3)
interpersonal trust	0.111 * (0.062)	0.200 * (0.117)	0.251 * (0.148)
institutional trust	0.354 *** (0.101)	0.628 *** (0.179)	0.869 *** (0.230)
control variables	controlled	controlled	controlled
Prob > chi^2	0.000	0.000	0.000
Pseudo R^2	0.159	0.160	0.204
log likelihood	−347.067	−346.665	−210.385
numbers of observation	701	701	435

Note: * and *** respectively represent significance levels of 10% and 1%. Robust standard errors are in parentheses.

4.5. Heterogeneity Analysis

Household endowment refers to the resources and capabilities that a farmer's household possesses, which have a significant impact on individual decision-making behavior [44]. Due to the heterogeneity of the farmer group, each farmer has different endowment characteristics, such as capital, labor, and education level. These differences may have varying impacts on their production activities. Therefore, this study not only explores the impact of social trust on farmers' agricultural waste recycling behavior, but also further analyzes whether the influence of social trust varies across different farmer groups with different household endowments. Referring to existing studies, the educational level, the number of household laborers, and the proportion of agricultural income are used as indicators to represent the cultural capital endowment, human capital endowment, and economic capital endowment of farmers [45]. The average values of these three indicators are used as grouping criteria to divide the farmer sample into two groups: "below the average group" and "above the average group", and separate regressions are conducted on each subgroup [46]. The estimation results are shown in Table 7.

In terms of educational level, the agricultural waste recycling behavior of farmers with lower education levels is significantly positively influenced by both interpersonal trust and institutional trust. In contrast, for farmers with higher education levels, their participation behavior is only significantly positively influenced by institutional trust, with interpersonal trust showing no significant effect. Therefore, it can be inferred that as farmers' educational levels increase, the influence of interpersonal trust on their participation in agricultural waste recycling gradually weakens, while the promoting effect of institutional trust remains significant. This may be because farmers with lower educational levels tend to be less proactive in independently learning about and participating in agricultural waste recycling. As a result, advice from family, friends, and village officials significantly impacts their production behavior. On the other hand, farmers with higher educational levels tend to have higher cognitive abilities and more attention to rural ecological environmental governance that directly affects their personal interests. As a result, institutional trust has a stronger promoting effect on agricultural waste recycling behavior among farmers with higher education levels.

Table 7. The results of heterogeneity analysis.

Variables	Educational Level		Numbers of Household Labor		Proportion of Agricultural Income	
	Below Mean Group	Above Mean Group	Below Mean Group	Above Mean Group	Below Mean Group	Above Mean Group
interpersonal trust	0.231 * (0.139)	0.216 (0.199)	0.057 (0.144)	0.412 ** (0.193)	0.188 (0.135)	0.111 (0.214)
institutional trust	0.384 * (0.219)	1.117 *** (0.322)	0.840 *** (0.237)	0.570 * (0.299)	0.566 *** (0.210)	0.866 ** (0.391)
control variables	controlled	controlled	controlled	controlled	controlled	controlled
Prob > chi^2	0.000	0.000	0.000	0.000	0.000	0.000
Pseudo R^2	0.171	0.253	0.161	0.235	0.221	0.285
log likelihood	−230.909	−99.607	−174.524	−149.054	−216.283	−96.124
numbers of observation	484	217	319	382	481	220

Note: *, **, and *** respectively represent significance levels of 10%, 5%, and 1%. Robust standard errors are in parentheses.

In terms of the number of family laborers, the participation of farm households with a smaller number of family laborers in agricultural waste resource utilization is only affected by the variable of institutional trust. The possible explanation is that for farm households with a smaller number of family laborers, farmers tend to exchange the family's effective labor inputs for higher comparative returns [47], and thus have a higher level of trust in village cadres as well as the government's guidance regarding agricultural production. Whereas institutional trust has a significant positive effect on both types of farmers' participation in agricultural waste resource utilization under different agricultural income share qualifications, interpersonal trust has no significant effect on the participation behavior of both types of farmers. This may suggest that for farmers, resource utilization of agricultural waste is equivalent to a long-term investment, and farmers are more in need of support at the institutional level.

5. Discussion

5.1. Similarities and Differences with Existing Research

This paper is consistent with previous research on agricultural waste resource utilization behavior to determine that interpersonal trust and institutional trust can positively influence farmers' participation behavior [28]. However, this paper extends this research by incorporating the variable of environmental regulation policy and constructing a comprehensive analytical framework to demonstrate the moderating effect of environmental regulation policy in social trust on farmers' agricultural waste resource utilization behavior. The moderating effect exists in the context of guided environmental regulation policy, which is consistent with Zhu et al.; however, the difference is that Zhu et al. found that the direction of the role of guided environmental regulation policy is positive [32], whereas in this paper, guided environmental regulation policy positively moderates the relationship between interpersonal trust-participation behavior and plays an alternative moderating role in institutional trust-participation behavior. The constrained environmental regulation policy can play a moderating effect between institutional trust and farmers' agricultural waste resource utilization behavior, which is consistent with Li and Wang's study [35]. In addition, this study supports the conclusion that incentive-based environmental regula-

tory policies do not play a moderating effect in the influence of social trust on farmers' agricultural waste resource utilization behavior [8].

5.2. Theoretical and Practical Implications

Based on the above theoretical analysis and empirical test, the research insights of this paper are as follows: (1) Theoretical implications. The findings of this paper clarify the role of social trust and environmental regulation policies in promoting farmers' participation in agricultural waste resource utilization, which helps to expand the research on the influencing factors and mechanisms of farmers' agricultural waste resource utilization behavior. (2) Practical implications. Currently, in the governance situation dominated by environmental regulation policies, the positive role of social trust in solving the rural ecological environment governance problems has been recognized, and integrating social trust into the decision-making of regulating the agricultural waste resource utilization behavior of farmers may become an important path to improve the rural ecological environment.

5.3. Limitation and Future Research Direction

The limitation of the research in this paper is the geographical homogeneity of the sample. Jiangsu Province is representative as a sample province for the study. However, given the vastness of China's geographic area and the significant regional variability in agricultural production, coupled with the different social, economic, and cultural environments in each region, it remains to be investigated whether the findings of this paper are applicable to other provinces. Therefore, in the future, we should expand the coverage of the research area, increase cross-regional comparative research, and demonstrate the decision-making process and mechanism of agricultural waste resource utilization behavior of farmers nationwide.

6. Conclusions and Recommendations

6.1. Conclusions

Based on micro-survey data from farmers in Jiangsu Province, this study uses a binary logistic regression model to explore how social trust influences farmers' agricultural waste recycling behavior. Additionally, it analyzes the moderating effect of environmental regulation policies on the relationship between social trust and participation behavior. The results show that both interpersonal trust and institutional trust effectively promote farmers' participation in agricultural waste recycling behavior. However, based on the marginal effect results, the impact of institutional trust is more significant. Secondly, different types of environmental regulation policies have differentiated effects on the role of interpersonal and institutional trust in promoting farmers' agricultural waste recycling behavior. Guiding environmental regulation policies positively moderate the influence of interpersonal trust on farmers' recycling behavior, while both guiding and constraining environmental regulation policies exhibit a substitutive moderating effect in the relationship between institutional trust and farmers' participation behavior. Finally, due to differences in family endowments, interpersonal and institutional trust have varying effects on agricultural waste recycling behavior across different farmer groups. As farmers' education level rises, family labor increases, and the proportion of agricultural income decreases, institutional trust continues to play a role in farmers' decision-making.

6.2. Policy Recommendations

The findings of this study provide the following three insights for promoting agricultural waste recycling and improving the agricultural ecological environment: First, emphasize the positive influence of social trust on farmers' decision-making in produc-

tion. Through rural community activities, cooperation and exchanges between farmers have been strengthened, and farmers have been motivated to participate in the resource utilization of agricultural waste. Additionally, public services at the village level should serve as a link to better connect village leaders with farmers, thereby promoting farmers' participation in agricultural waste recycling and creating a supportive social environment for its implementation. Second, coordinate the roles of social trust and environmental regulation policies in promoting farmers' participation in agricultural waste recycling. Continuously improve environmental regulation policies that align with rural development and agricultural production levels, increase the promotion and dissemination of these policies, and meet farmers' needs in production and operations. At the same time, within the scope of environmental regulation policies, increase farmers' understanding and trust in agricultural waste recycling, and strengthen the interaction between the two. Third, when promoting agricultural waste recycling and related policies, it is important to fully consider the heterogeneity of individual farmers and their families. For farmers with lower education levels, larger family labor forces, and higher agricultural income shares, the role of new business entities such as professional cooperatives should be fully utilized, with agricultural technicians providing specific production guidance to enhance farmers' social trust and improve their understanding and acceptance of agricultural waste recycling. For others, a public service mechanism involving multiple stakeholders, including the government, social organizations, and farmers, should be established to further leverage institutional trust in promoting farmers' agricultural waste recycling behaviors.

Author Contributions: Conceptualization, J.W.; methodology, L.M.; software, L.M.; validation, J.W.; formal analysis, L.M.; investigation, J.W.; data curation, L.M.; writing—original draft preparation, L.M.; writing—review and editing, J.W.; supervision, J.W.; project administration, J.W.; funding acquisition, J.W. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by The National Social Science Fund of China (grant number: 20&ZD117).

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the authors.

Acknowledgments: The authors thank the participants for their contributions to this research.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Wu, S.; Xing, L.; Wang, J.; Jia, M.; Liu, C.; Zhou, Q.; Wang, L. Domestic and Foreign Research Hotspots and Development Trend of Agricultural Waste Resource Utilization. *Chin. Agric. Sci. Bull.* **2024**, *40*, 148–156.
2. Koul, B.; Yakoob, M.; Shah, M.P. Agricultural Waste Management Strategies for Environmental Sustainability. *Environ. Res.* **2022**, *206*, 112285. [CrossRef] [PubMed]
3. Han, J.; He, X. Development of circular economy is a fundamental way to achieve agriculture sustainable development in China. *Energy Procedia* **2011**, *5*, 1530–1534. [CrossRef]
4. Velasco-Muñoz, J.; Mendoza, J.; Aznar-Sánchez, J.; Gallego-Schmid, A. Circular economy implementation in the agricultural sector: Definition, strategies and indicators. *Resour. Conserv. Recycl.* **2021**, *170*, 105618. [CrossRef]
5. Wang, J.; Tao, J.; Chen, L. Resource Utilization Method and Influencing Factors of Farmers' Behaviors towards Livestock and Poultry Waste. *China Popul. Resour. Environ.* **2019**, *29*, 127–137. [CrossRef]
6. Li, A.; Li, X. Study on Farmer's Willingness and Behavior about Agricultural Waste Recycling Based on the Theory of Planned Behavior. *J. Arid Land Res. Environ.* **2019**, *33*, 33–40. [CrossRef]
7. Kassie, M.; Jaleta, M.; Shiferaw, B.; Mmbando, F.; Mekuria, M. Adoption of Interrelated Sustainable Agricultural Practices in Smallholder Systems: Evidence from Rural Tanzania. *Technol. Forecast. Soc. Change* **2013**, *80*, 525–540. [CrossRef]

8. Wang, J.; Dou, L.; Wang, Y. The Impact of Agricultural Marketization on Livestock Waste Resource Utilization in the Context of Environmental Regulation Policy. *Chin. Rural Econ.* **2022**, *1*, 93–111.
9. Wang, H.; Xiao, L.; Huang, Y. Farmers' Ecological Environmental Protection Awareness and Its Impact on Wastes Utilization. *J. Agric. For. Econ. Manag.* **2020**, *19*, 699–706. [CrossRef]
10. Langenbach, B.P.; Berger, S.; Baumgartner, T.; Knoch, D. Cognitive Resources Moderate the Relationship Between Pro-Environmental Attitudes and Green Behavior. *Environ. Behav.* **2020**, *52*, 979–995. [CrossRef]
11. Pan, D.; Kong, F. An Analysis of Waste Cyclic Utilization in Livestock Breeding Based on the Grounded Theory: Farmers' Behavior and Policy Intervention Paths. *J. Jiangxi Univ. Finance Econ.* **2018**, *3*, 95–104. [CrossRef]
12. Wang, J.; Dou, L.; Ma, L. Analysis of the Driving Mechanism for Farmers' Integration into Agricultural Green Production Transformation: A Case Study of Agricultural Waste Resource Utilization. *J. Nanjing Agric. Univ. (Soc. Sci. Ed.)* **2023**, *23*, 165–177. [CrossRef]
13. Shi, H.; Sui, D.; Wu, H.; Zhao, M. The Influence of Social Capital on Farmers' Participation in Watershed Ecological Management Behavior: Evidence from Heihe Basin. *Chin. Rural Econ.* **2018**, *1*, 34–45. [CrossRef]
14. Yang, L.; Zhu, Y. Effects of Social Trust and Cooperative Ability on Farmers' Participation Behavior in Supply of Small-scale Farmland Water Conservancy: Based on the Data of Five Provinces in Yellow River Irrigation District. *China Popul. Resour. Environ.* **2016**, *26*, 163–170. [CrossRef]
15. Wang, W.; Zhao, X.; Li, H.; Zhang, Q. Will Social Capital Affect Farmers' Choices of Climate Change Adaptation Strategies? Evidence from Rural Households in the Qinghai-Tibetan Plateau, China. *J. Rural Stud.* **2021**, *83*, 127–137. [CrossRef]
16. Wang, P.; Li, G.; Liu, D.; Gao, Y. Influence of Agricultural Extension Adoption of Modes on Farmers' Scientific Fertilization Techniques: The Moderating Effect of Social Trust. *J. Arid Land Res. Environ.* **2024**, *38*, 150–157. [CrossRef]
17. Ye, Z.; Zhong, Z. Benefit Expectation, Social Trust and Farmer's Choice of Farmland Transfer-out Contract. *Arid Land Res. Environ.* **2022**, *36*, 72–80. [CrossRef]
18. Li, X.; Liu, J.; Huo, X. Impact of Social Trust on Farmland Rental Market. *J. Nanjing Agric. Univ. (Soc. Sci. Ed.)* **2020**, *20*, 128–139. [CrossRef]
19. Ren, Z.; Guo, Y. The Effect of Environmental regulation and Social Capital on Farmers' Adoption Behavior of Low-carbon Agricultural Technology. *J. Nat. Resour.* **2023**, *38*, 2872–2888. [CrossRef]
20. Zhang, B.; Fu, S. How Do Agricultural Policies Influence Farmers' Decisions on the Resource Utilization of Productive Agricultural Waste? An Empirical Study Based on 1227 Farm Household Survey Data from Anhui Province. *Rural Econ.* **2023**, *9*, 98–108.
21. Mueller, W. The Effectiveness of Recycling Policy Options: Waste Diversion or Just Diversions? *Waste Manag.* **2013**, *33*, 508–518. [CrossRef] [PubMed]
22. Kim, J.; Goldsmith, P.; Thomas, M.H. Economic Impact and Public Costs of Confined Animal Feeding Operations at the Parcel Level of Craven County, North Carolina. *Agric. Hum. Values* **2010**, *27*, 29–42. [CrossRef]
23. Zhao, Y.; Zhou, H. Can Social Trust and Policy of Rewards and Punishments Promote Farmers' Participation in the Recycling of Pesticide Packaging Waste? *J. Arid Land Res. Environ.* **2021**, *35*, 17–23. [CrossRef]
24. Li, F.; Zhang, J.; He, K. Alternative and Complementary: Informal Institutions and Formal Institutions in Farmers' Green Production. *J. Huazhong Univ. Sci. Technol. (Soc. Sci. Ed.)* **2019**, *33*, 51–60+94. [CrossRef]
25. Delhey, J.; Newton, K.; Welzel, C. How General is Trust in "Most People"? Solving the Radius of Trust Problem. *Am. Sociol. Rev.* **2011**, *76*, 786–807. [CrossRef]
26. Dinesen, P.T.; Sønderskov, K.M. Ethnic Diversity and Social Trust: Evidence from the Micro-Context. *Am. Sociol. Rev.* **2015**, *80*, 550–573. [CrossRef]
27. Tsang, S.; Burnett, M.; Hills, P.; Welford, R. Trust, Public Participation, and Environmental Governance in Hong Kong. *Environ. Policy Gov.* **2009**, *19*, 99–114. [CrossRef]
28. He, K.; Zhang, J.; Zhang, L.; Wu, X. Interpersonal Trust, Institutional Trust, and Farmers' Willingness to Participate in Environmental Governance: A Case of Agricultural Waste Recycling. *Manag. World* **2015**, *5*, 75–88. [CrossRef]
29. Li, Q.; Wang, Y. On the Choices of Government Behaviors in the Utilization of Livestock and Poultry Waste Resources. *Rural Econ.* **2018**, *9*, 55–61.
30. Yu, T.; Yu, F. The Impact of Cognition of Livestock Waste Resource Utilization on Farmers' Participation Willingness in the Context of Environmental Regulation Policy. *Chin. Rural Econ.* **2019**, *8*, 91–108. [CrossRef]
31. Si, R.; Lu, Q.; Zhang, S. Effect of Environmental Regulation on Household Dead Pig Recycling Disposal Behavior: Based on the Empirical Data in Hebei, Henan, and Hubei Province. *J. Agric. Econ.* **2020**, *7*, 47–60. [CrossRef]
32. Zhu, R.; He, K.; Zhang, J. How Do Environmental Regulations Affect Farmers' Decision-making of Utilizing Livestock and Poultry Manure as Resources? From the Perspective of Perceptions of Large-scale Pig Farmers. *China Rural Surv.* **2021**, *6*, 85–107. [CrossRef]
33. He, K.; Zhang, J. The Ecological Value of Agricultural Waste Recycling: A Comparative Analysis of the Willingness to Pay between New Generation and Previous Generation Farmers. *China Rural Econ.* **2014**, *5*, 62–73+85. [CrossRef]

34. Pan, D.; Kong, F. Analysis of Livestock Farmers' Choices for Environmentally Friendly Animal Manure Disposal: A Case Study of Pig Farming. *China Rural Econ.* **2015**, *9*, 17–29. [CrossRef]
35. Li, W.; Wang, G. Influence of Social Capital on Farmers' Adoption of Breeding Waste Recycling Technology: Regulatory Role of Environmental Regulation Policy. *J. Agric. For. Econ. Manag.* **2021**, *20*, 199–208. [CrossRef]
36. Zhang, T.; Yan, T.; Zhang, J. How the Rural Cadre-farmer Relationship Affects the Farmers' Agricultural Waste Utilization: Based on the Survey Data of 1372 Farmers in Four Provinces. *J. Nanjing Agric. Univ. (Soc. Sci. Ed.)* **2020**, *20*, 150–160. [CrossRef]
37. Yan, T.; He, K.; Zhang, J. Analysis of Social Capital Influencing Farmers' Willingness of Environmental Protection Investment: Evidence from Empirical Study on Reusing Agricultural Wastes in Hubei Rural Areas. *China Popul. Resour. Environ.* **2016**, *26*, 158–164. [CrossRef]
38. Bouma, J.; Bulte, E.; Van, S.D. Trust and Cooperation: Social Capital and Community Resource Management. *J. Environ. Econ. Manag.* **2008**, *56*, 155–166. [CrossRef]
39. Tang, L.; Luo, X.; Zhang, J. Social Supervision, Group Identity and Farmers' Domestic Waste Centralized Disposal Behavior: An Analysis Based on Mediation Effect and Regulation Effect of the Face Concept. *China Rural Surv.* **2019**, *2*, 18–33. [CrossRef]
40. Zhang, Y.; Jiang, Y. The Impact of Environmental Risk Perception on Pig Farmers' Environmental Behavior under Environmental Regulation Policy Context: A Survey of 280 Large-Scale Farmers in Hubei Province. *Agric. Technol. Econ.* **2016**, *11*, 76–86. [CrossRef]
41. Yang, J.; Zhang, W.; Zhou, L. The Complementarity and Substitutability of Reputation and Regulation: Empirical Study Based on Online Transaction Data. *Manag. World* **2008**, *7*, 9. [CrossRef]
42. Li, F.; Zhang, J.; He, K. lmpact of Informal Institutions and Environmental Regulations on Farmers' Green Production Behavior: Based on Survey Data of 1105 Households in Hubei Province. *Res. Sci.* **2019**, *41*, 1227–1239. [CrossRef]
43. Yang, Z. Ageing, Social Network and the Adoption of Green Production Technology: Evidence from Farm Households in Six Provinces in the Yangtze River Basin. *China Rural Surv.* **2018**, *4*, 44–58. [CrossRef]
44. Yuan, T.; Zong, Y.; Wang, J. Farmers' Adoption Behavior of Organic Soil Improvement Technology: External Incentives and Endogenous Driving. *Agric. Technol. Econ.* **2021**, *8*, 92–104. [CrossRef]
45. Zhang, C.; Li, Y.; Li, H. Capital Endowment and Farmers' Participation Behavior in Social Governance: An Empirical Analysis Based on Data from 1599 Farmers in Five Provinces. *China Rural Surv.* **2016**, *1*, 27–37+50. [CrossRef]
46. Zhang, W.; Wang, D.; Lu, Q. Impact of Agricultural Productive Services on Green Production Behavior of Farmers. *J. Arid Land Res. Environ.* **2024**, *38*, 21–30. [CrossRef]
47. Zhao, K.; Zhang, R.; Sun, P. The Impacts of Capital Endowment on Farmers' Adoption Behaviors of Agricultural Socialization Services: From the Perspective of Family Life Cycle. *Agric. Mod. Res.* **2022**, *43*, 121–133. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.